

S M (Rifat) Rafiuddin

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EDUCATION

Oklahoma State University

Stillwater, OK

Ph.D. in Computer Science

August 2022 – Expected July 2027

- **Advisor:** Dr. Atriya Sen
- **Provisional Dissertation:** *Efficient, Interpretable, and Controllable Language Models through Representation-Guided Adaptation and Intervention*

Rajshahi University of Engineering and Technology

Rajshahi, Bangladesh

B.Sc. in Computer Science and Engineering

January 2012 – October 2016

ACADEMIC APPOINTMENTS

Oklahoma State University

Stillwater, OK

Graduate Teaching and Research Assistant, Department of Computer Science

August 2022 – Present

- **Graduate Teaching Assistant:** Academic-year appointments, August–May, beginning Fall 2022.
- **Graduate Research Assistant, Complex Systems Lab:** Summers 2023 and 2024.
- **Graduate Research Assistant, Reasoning and Artificial Intelligence (rAIson) Lab:** Summers 2025 and 2026.

University of Asia Pacific

Dhaka, Bangladesh

Lecturer, Department of Computer Science and Engineering

October 2018 – July 2022

Uttara University

Dhaka, Bangladesh

Lecturer, Department of Computer Science and Engineering

February 2017 – October 2018

RESEARCH INTERESTS

Natural language processing and representation learning, with emphasis on efficient and interpretable language models, parameter-efficient fine-tuning, mechanistic analysis, causal and counterfactual reasoning, and targeted interventions for improving model faithfulness, reliability, and control.

PUBLICATIONS

Publications are grouped by type and listed in descending order by publication year.

Refereed Conference and Findings Papers

- [1] **S M Rafiuddin** and Rafae Abdullah. “Context-Conditioned Masked LoRA: Dynamic Rank Routing for Compute-Efficient Parameter-Efficient Fine-Tuning.” In *Findings of the Association for Computational Linguistics: ACL 2026*, pp. 26670–26689. Association for Computational Linguistics, 2026. [**Paper**]
- [2] **S M Rafiuddin** and Muntaha Nujat Khan. “Emergent Discrete Controller Modules for Symbolic Planning in Transformers.” In *International Conference on Learning Representations (ICLR 2026)*, Rio de Janeiro, Brazil, 2026. [**Paper**] [**Presentation**] [**Poster**] [**Video**]
- [3] **S M Rafiuddin**. “MaskLoRA: Low-Rank Subspace-Induced Token Masking for Efficient and Faithful Language Models.” In *Findings of the Association for Computational Linguistics: EACL 2026*, pp. 5678–5692. Association for Computational Linguistics, 2026. [**Paper**] [**Presentation**] [**Poster**] [**Video**]
- [4] **S M Rafiuddin** and Muntaha Nujat Khan. “A Formal Analysis of Chain-of-Thought Prompting via Turing Reductions.” In *Proceedings of the 14th International Joint Conference on Natural Language Processing and the 4th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics (IJCNLP–AACL 2025): Short Papers*, pp. 87–98. Asian Federation of Natural Language Processing and Association for Computational Linguistics, 2025. [**Paper**]
- [5] Sadia Kamal, Lalu Prasad Yadav Prakash, **S M Rafiuddin**, Mohammed Rakib, Atriya Sen, and Sagnik Ray Choudhury. “A Detailed Factor Analysis for the Political Compass Test: Navigating Ideologies of Large

- Language Models.” In *Proceedings of the 14th International Joint Conference on Natural Language Processing and the 4th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics (IJCNLP–AAACL 2025): Short Papers*, pp. 284–303. Asian Federation of Natural Language Processing and Association for Computational Linguistics, 2025. [Paper]
- [6] **S M Rafiuddin**. “Edu-EmotionNet: Cross-Modality Attention Alignment with Temporal Feedback Loops.” In *Proceedings of the 24th International Conference on Machine Learning and Applications (ICMLA 2025)*, pp. 200–205. IEEE, 2025. [Paper] [Presentation]
- [7] Md. Sohanuzzaman Soad, Mahady Al Hady, **S M Rafiuddin**, and Sudip Ghose. “Cross-Domain Adversarial Augmentation: Stabilizing GANs for Medical and Handwriting Data Scarcity.” In *Proceedings of the 2025 IEEE 4th International Conference on Robotics, Automation, Artificial-Intelligence and Internet-of-Things (RAAICON 2025)*. IEEE, 2025. [Paper]
- [8] **S M Rafiuddin** and Muntaha Nujat Khan. “Learning What to Remember: Adaptive Probabilistic Memory Retention for Memory-Efficient Language Models.” In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pp. 3969–3981. Association for Computational Linguistics, 2025. [Paper]
- [9] **S M Rafiuddin**, Sadia Kamal, Mohammed Rakib, Arunkumar Bagavathi, and Atriya Sen. “AdaptiSent: Context-Aware Adaptive Attention for Multimodal Aspect-Based Sentiment Analysis.” In *Social Networks Analysis and Mining: Proceedings of the 17th International Conference on Advances in Social Networks Analysis and Mining (ASONAM 2025)*, Part I, Lecture Notes in Computer Science, vol. 16322, pp. 133–141. Springer, 2025. [Paper] [Presentation]
- [10] **S M Rafiuddin**, Mohammed Rakib, Sadia Kamal, and Arunkumar Bagavathi. “Exploiting Adaptive Contextual Masking for Aspect-Based Sentiment Analysis.” In *Advances in Knowledge Discovery and Data Mining: Proceedings of the 28th Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD 2024)*, Part VI, Lecture Notes in Computer Science, vol. 14650, pp. 147–159. Springer, 2024. [Paper] [Presentation]
- [11] Md. Abdul Karim, **S M Rafiuddin**, Md. Jahidul Islam Razin, and Tahira Alam. “Isolated Bangla Handwritten Character Classification Using Transfer Learning.” In *Proceedings of the 2nd International Conference on Computing Advancements (ICCA 2022)*, pp. 11–17. Association for Computing Machinery, 2022. [Paper]
- [12] **S M Rafiuddin**. “High Cursive Complex Character Recognition Using GAN External Classifier.” In *Proceedings of the 2nd International Conference on Computing Advancements (ICCA 2022)*, pp. 466–472. Association for Computing Machinery, 2022. [Paper]
- [13] Md. Jahidul Islam Razin, Md. Abdul Karim, M. F. Mridha, **S M Rafiuddin**, and Tahira Alam. “A Long Short-Term Memory (LSTM) Model for Business Sentiment Analysis Based on Recurrent Neural Network.” In *Sustainable Communication Networks and Application*, Lecture Notes on Data Engineering and Communications Technologies, vol. 55, pp. 1–15. Springer, 2021. [Paper]
- [14] **S M Rafiuddin**. “Estimation of Phylogenetic Tree Using Gene Sequencing Data.” In *2019 4th International Conference on Electrical Information and Communication Technology (EICT)*, pp. 1–5. IEEE, 2019. [Paper]
- [15] **S M Rafiuddin**. “Ranking of Bangla Word Graph Using Graph-Based Ranking Algorithms.” In *2017 3rd International Conference on Electrical Information and Communication Technology (EICT)*, pp. 1–5. IEEE, 2017. [Paper]
- [16] Sadia Zaman Mishu and **S M Rafiuddin**. “Performance Analysis of Supervised Machine Learning Algorithms for Text Classification.” In *2016 19th International Conference on Computer and Information Technology (ICCIT)*, pp. 409–413. IEEE, 2016. [Paper]

Refereed Workshop Papers

- [17] **S M Rafiuddin**, Vamsi Krishna Pavuluri, and Atriya Sen. “Constraint-Aware Counterfactual Editing for Aspect-Based Sentiment Analysis.” Presented at the *2nd International Workshop on Informing Machine Learning with Knowledge Engineering for Hybrid Intelligent Systems (HHAI-KEML 2026)*, Brussels, Belgium, July 6–7, 2026. [Paper]

SELECTED RESEARCH PROJECTS AND SOFTWARE

- Estimating Influenza Cases Using Numerical Methods and Machine Learning** [Code] 2024
- Compared SVIR-based midpoint and fourth-order Runge–Kutta solvers with LSTM, CNN, FTA-LSTM, and Random Forest models for forecasting H1N1 outbreaks; Random Forest achieved the strongest predictive performance.
- Image Embedding and Classification with Deep Neural Networks** [Code] [Paper] 2021
- Developed an Xception-based classification and representation-analysis pipeline with TensorBoard visualizations of learned embeddings in two and three dimensions.
- Data Augmentation with Generative Adversarial Networks** [Code] [Demo] 2021
- Implemented adaptive discriminator augmentation to stabilize GAN training in limited-data settings, supporting Bangla handwritten character and numeral synthesis across multiple architectures.
- Phylogenetic Tree Construction with Genetic Algorithms** [Code] [Paper] 2019
- Developed a genetic-algorithm framework for estimating phylogenetic trees from gene-sequencing data, with modular support for multiple genomic datasets from the National Center for Biotechnology Information.

RESEARCH EXPERIENCE

- Reasoning and Artificial Intelligence (rAIsen) Lab** Oklahoma State University
Graduate Research Assistant; Advisor: Dr. Atriya Sen Summers 2025 and 2026
- Developed representation-guided and parameter-efficient adaptation methods for language models, including token masking, low-rank subspace intervention, and context-conditioned rank routing.
 - Investigated the internal representations and computational structures underlying transformer reasoning, symbolic planning, and model faithfulness.
 - Designed constraint-aware counterfactual editing methods for aspect-based sentiment analysis, emphasizing target-aspect intervention, non-target preservation, semantic consistency, and minimal editing.
 - Built reproducible PyTorch and Hugging Face experimentation pipelines incorporating controlled ablations, memory and throughput profiling, and comparative evaluation.
- Complex Systems Lab** Oklahoma State University
Graduate Research Assistant; Advisor: Dr. Arunkumar Bagavathi Summers 2023 and 2024
- Developed adaptive contextual masking methods for aspect-based sentiment analysis to improve aspect-specific representation learning and sentiment prediction.
 - Investigated multimodal aspect-based sentiment analysis using context-aware attention and text–image representation alignment.
 - Conducted model comparisons, component ablations, and error analyses to evaluate the contributions of masking, attention, and multimodal fusion mechanisms.

RESEARCH SUPPORT AND AWARDS

- Research and Travel Support**
- ICLR 2026 Financial Assistance Award**, International Conference on Learning Representations 2026
- Awarded complimentary registration and \$1,600 in travel support to present research at ICLR 2026 in Rio de Janeiro, Brazil.
- AIRS Travel Fund**, Oklahoma State University 2025
- Received institutional support for presenting and disseminating peer-reviewed research.
- GPSGA Travel and Research Award**, Oklahoma State University 2024–2025
- Awarded \$600 in 2024 and \$300 in 2025 to support conference travel and research dissemination.
- Selected Competitive Honors**
- Honorable Mention, ICT Fest**, Islamic University of Technology 2014
- Honorable Mention, National Collegiate Programming Contest**, Daffodil International University 2014
- Champion, ICT Olympiad – CSE Fest**, Rajshahi University of Engineering and Technology 2012

TEACHING EXPERIENCE

Oklahoma State University

Graduate Teaching Assistant, Department of Computer Science

Stillwater, OK

August 2022 – Present

- **Introduction to Computer Security** — Fall 2022; Instructor: Dr. Sharmin Jahan.
- **Design and Implementation of Operating Systems I** — Spring 2023, Spring 2024, and Spring 2026; Instructor: Dr. Shital Joshi.
- **Data Structures and Algorithm Analysis II** — Fall 2023; Instructor: Dr. H. K. Dai.
- **Discrete Mathematics for Computer Science** — Fall 2024; Instructor: Dr. Sachin Jain.
- **Social Issues in Computing** — Spring 2025; Instructor: Dr. Rekha Bhowmik.
- **Introduction to Database Systems** — Fall 2025; Instructor: Dr. Sadiq Albuhamood.
- Led laboratory and discussion sessions, graded programming assignments and examinations, provided individual student support, and assisted with course administration and instructional-material development.

University of Asia Pacific

Lecturer and Instructor of Record, Department of Computer Science and Engineering

Dhaka, Bangladesh

October 2018 – July 2022

- **Theory Courses:** Machine Learning; Pattern Recognition; Design and Analysis of Algorithms; Mathematics for Computer Science; and Visual and Web Programming.
- **Laboratory Courses:** Computer Graphics; Pattern Recognition; Compiler Design; Algorithms; Object-Oriented Programming II (Java); and Visual and Web Programming.
- Designed and delivered lectures, laboratory exercises, assignments, examinations, and grading rubrics; supervised undergraduate projects and contributed to outcome-based curriculum development.

Uttara University

Lecturer and Instructor of Record, Department of Computer Science and Engineering February 2017 – October 2018

Dhaka, Bangladesh

- **Courses Taught:** Design and Analysis of Algorithms; Operating System Design; Discrete Mathematics; and Programming Language and Application II (C++).
- Delivered undergraduate lectures and laboratory sessions, prepared instructional materials and examinations, evaluated student work, and supported departmental assessment and result preparation.

STUDENT RESEARCH MENTORING

Reasoning and Artificial Intelligence (rAIson) Lab

Research Mentor; Faculty Supervisor: Dr. Atriya Sen

Oklahoma State University

Spring 2026

- **Vamsi Krishna Pavuluri — Constraint-Aware Counterfactual Editing for Aspect-Based Sentiment Analysis:** Mentored research on counterfactual generation, aspect-specific sentiment intervention, semantic preservation, experimental evaluation, and manuscript preparation.
- The project resulted in the co-authored peer-reviewed paper “Constraint-Aware Counterfactual Editing for Aspect-Based Sentiment Analysis,” presented at the 2nd International Workshop on Informing Machine Learning with Knowledge Engineering for Hybrid Intelligent Systems (HHAI-KEML 2026), Brussels, Belgium.

Complex Systems Lab

Research Mentor

Oklahoma State University

Fall 2023

- **Jake Williams — Multimodal Aspect-Based Sentiment Analysis:** Mentored research on multimodal representation learning, aspect-level sentiment modeling, experimental design, model evaluation, and technical communication.

University of Asia Pacific

Student Research Mentor, Department of Computer Science and Engineering

Dhaka, Bangladesh

2018 – 2022

- **Md. Sohanuzzaman Soad, Mahady Al Hady, and Sudip Ghose — Cross-Domain Adversarial Data Augmentation:** Mentored research on GAN-based data augmentation for medical and handwriting datasets, including problem formulation, method development, experimental evaluation, and manuscript preparation.
- The project resulted in the co-authored peer-reviewed paper “Cross-Domain Adversarial Augmentation: Stabilizing GANs for Medical and Handwriting Data Scarcity,” presented at IEEE RAAICON 2025.

RESEARCH COLLOQUIUM PRESENTATIONS

“Multimodal Representation Learning.” Joint Ph.D. Research Colloquium, Departments of Computer Science, Oklahoma State University and the University of Tulsa. Spring 2025

- Surveyed joint, coordinated, contrastive, and transformer-based approaches to multimodal representation learning; discussed missing and imbalanced modalities and presented an adaptive modality-weighting framework as a prospective research direction.

“Literature Review on Aspect-Based Sentiment Analysis.” Joint Ph.D. Research Colloquium, Departments of Computer Science, Oklahoma State University and the University of Tulsa. Spring 2023

- Reviewed major developments in aspect extraction and sentiment classification, including pretrained language models, syntactic and graph-based methods, multitask learning, and unified generative formulations for aspect-based sentiment analysis.

“ECA-GAN: Low-Sampled Classification Using Semi-Supervised GAN Embedding with an External Classifier.” Joint Ph.D. Research Colloquium, Departments of Computer Science, Oklahoma State University and the University of Tulsa. Fall 2022

- Presented a semi-supervised framework combining GAN-based data augmentation, pseudo-labeling, embedding-space analysis, and an external classifier for image classification under limited labeled-data conditions.

PROFESSIONAL SERVICE

Peer Review

ACM Transactions on Internet Technology	2026
Conference on Neural Information Processing Systems (NeurIPS), Position Paper Track	2026
International AAAI Conference on Web and Social Media (ICWSM)	2026
Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD)	2025
International Conference on Computational Linguistics (COLING)	2024
International Joint Conference on Neural Networks (IJCNN)	2024

Academic Outreach and Community Service

Competitive Programming Team Coach	University of Asia Pacific; RUET Inter-University Programming Contest	2019
Volunteer	National High School Programming Contest, Rajshahi	2016
Mathematics Olympiad Volunteer	Divisional Mathematical Olympiad, Faridpur	2006

FACULTY DEVELOPMENT AND TRAINING

Improving Learning and Teaching Skills (ILTS) , University of Asia Pacific	May 5, 2019
Roles, Responsibilities, and Ethical Principles of University Teachers , Institutional Quality Assurance Cell, Uttara University	February 24, 2018

TECHNICAL EXPERTISE

Research Areas and Methods	Language-model representation analysis, parameter-efficient fine-tuning, low-rank adaptation (LoRA), representation-guided token masking, mechanistic interpretability, causal and counterfactual intervention, aspect-based sentiment analysis, and multimodal representation learning
Frameworks and Libraries	PyTorch, Hugging Face Transformers, TensorFlow, Scikit-learn, NumPy, pandas, and Matplotlib
Programming Languages	Python, C, C++, Java, and SQL
Research Computing	Linux, Git, Docker, Amazon Web Services, GPU-accelerated experimentation, memory profiling, and throughput benchmarking
Data and Documentation	MySQL, Apache Spark, Jupyter, and L ^A T _E X

PROFESSIONAL MEMBERSHIP

Association for Computational Linguistics (ACL)	Member
<i>Academic references available upon request.</i>	